Roee Tekoah

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Computer science researcher and B.Sc. candidate at the Hebrew University of Jerusalem (expected 2027), with a peer-reviewed publication and **5+ years of technical work** across algorithms, computational biology, security, and systems.

EXPERIENCE

- **JSOF** · Security Researcher · Apr 2022–Jan 2024

  Black-box vulnerability research and attack-surface analysis of automotive and embedded platforms for intelligence agencies and commercial clients

  IDA Pro / IDAPython · ARM · Android · Automotive ECUs · Black-box RE

  - **ECU platform.** Led black-box offensive analysis of a complex automotive ECU, reverse engineering Android, custom-architecture, and proprietary-OS subsystems from partial, corrupted memory dumps
  - **Vulnerability discovery.** Identified and validated high-severity vulnerabilities—including logic flaws, authentication bypasses, and memory corruption—before customer deployment; collaborated with engineering, exploitation, and intelligence teams to guide remediation and support downstream defensive and offensive operations
  - **Tooling.** Built custom IDAPython tooling and reverse-engineering workflows to uncover exploitable code paths in obfuscated binaries

- **IDF (Matzov)** · Malware Researcher · Apr 2019–Apr 2022

  Large-scale malware research, collection, and detection for intelligence and defense teams

  C/C++ · Go · .NET · Python · Shellcode · YARA / Snort / Sigma · Memory forensics

  - **Malware analysis.** Reverse engineered heavily obfuscated, multi-stage malware using static and dynamic analysis, memory forensics, and dump analysis
  - **Live-sample hub.** Operated the intelligence community's central live-sample hub at national scale; expanded sample and IOC coverage through manual extraction, honeypot integration, and pipeline refinement
  - **Detection and attribution.** Produced IOCs, YARA/Snort/Sigma signatures, and threat reports; automated workflows supporting detection and attribution across intelligence and defense teams
  - **Recognition.** Key contributor to a team project awarded the Israel Defense Prize

- **Mobileye** · Software Engineer · Jun 2018–Mar 2019

  Algorithms for a pipeline transforming driving video into centimeter-level HD maps

  Python · NumPy / Pandas · Spark

  - **HD-map pipeline.** Developed geometric and signal-smoothing algorithms to reduce noise in HD-map overlays; built automation that reduced manual review and accelerated map validation
  - **Data pipelines.** Optimized large-scale graph and geometry pipelines using NumPy/Pandas and Spark

PUBLICATIONS & PROJECTS

GraFT – Alignment-Free HGT Detection · github.com/roeetekoah/GraFT · 2025

Graph-based detection over ~**324K proteins** (50 species); pruned 911K similarity edges to 92K, scored HGT candidates via k-mer Jaccard similarity, species-pair normalization (median/MAD), and graph-structural anomalies

D+ / X+ – X-ray Scattering Modeling · Raviv Research Group

Research Assistant · Peer-reviewed publication · J. Appl. Crystallogr. · 2019

Co-developed C++/C# analysis algorithms, including polymer-structure modules; built a geometric repeating-subunit detector (~**10**× **faster** at equal precision) and contributed to an octree method for protein hydration-shell detection

A. Ginsburg, T. Ben-Nun, R. Asor, A. Shemesh, L. Fink, **R. Tekoah** et al. *D+: software for high-resolution hierarchical modeling of solution X-ray scattering from complex structures*. *J. Appl. Crystallogr.*, 52:219–242

EDUCATION

The Hebrew University of Jerusalem
B.Sc. in Computer Science
2023–Present · Expected 2027

Gamma Cyber Program (Matzov), IDF
Selective Cyber Defense Program
2019–2020 · GPA 95

OS internals, network security, vulnerability research, reverse engineering, exploit analysis, malware analysis, memory forensics

SELECTED COURSEWORK

Complexity Theory **93**
Machine Learning **93**
Natural Language Processing **100**
Image Processing **97**
Computational Genomics **98**

SKILLS

Algorithms & Research

Graph algorithms · Computational geometry · Bioinformatics · Computational genomics · X-ray scattering

Programming & Data

C / C++ · Python · C# · Go · IDAPython · NumPy / Pandas · Spark

AI & Data

Machine learning · Natural language processing · Image processing · Large-scale data pipelines

Reverse Engineering

IDA Pro · Ghidra · Binary Ninja · GDB · Frida · Volatility · Wireshark · ProcMon / ProcDump · Assembly (x86/x64, ARM)

RESEARCH INTERESTS

Algorithms, computational biology, machine learning, complexity theory, security